

Google Form Creation Guide: User Requirements Questionnaire for Machine Learning-Based Cardiopulmonary Sound Separation System

Manual Google Form setup guide for Chapter 3 Requirements Analysis

Project	Machine Learning-Based System for Cardiopulmonary Sound Separation
Purpose	Collect practical user requirements for the proposed prototype.
Estimated time	3-5 minutes for respondents.

This guide is for manual form creation only. It does not create, distribute, or submit a Google Form online.

Purpose of the Questionnaire

The questionnaire is required for Chapter 3 Requirements Analysis and will be used to collect practical user requirements for the proposed prototype. It focuses on user background, current challenges, preferred features, interface design, output presentation, and evaluation needs.

Target Respondents

- Software Engineering / Computer Science students
- AI / Machine Learning students
- Biomedical Engineering / health-related students
- Researchers or students interested in biomedical audio or signal processing
- Supervisors/evaluators if appropriate

Google Form Title

Title	User Requirements Questionnaire for Machine Learning-Based Cardiopulmonary Sound Separation System
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Form Description / Introduction

Welcome and thank you for participating in this questionnaire.

This questionnaire is conducted as part of a Final Year Project at Multimedia University (MMU) involving the development of a machine learning-based system for cardiopulmonary sound separation. The proposed system aims to allow users to upload a mixed cardiopulmonary audio recording and obtain separated heart and lung sound outputs.

The purpose of this questionnaire is to understand user background, current challenges in working with biomedical or audio-processing tasks, and user preferences regarding system features, interface design, output presentation, and evaluation needs.

Your responses will help identify user requirements and support the design of the proposed prototype. There are no right or wrong answers. All responses will be used for academic purposes only and will remain confidential.

Estimated completion time: 3-5 minutes.

Manual Google Forms Steps

Step 1	Go to https://forms.google.com and click Blank Form.
Step 2	Set the form title: User Requirements Questionnaire for Machine Learning-Based Cardiopulmonary Sound Separation System
Step 3	Paste the introduction text into the form description.
Step 4	Create these sections: Section A - Respondent Background; Section B - Current Challenges in Cardiopulmonary Audio Processing; Section C - System Feature Preferences; Section D - Interface and Output Preferences; Section E - Evaluation and Improvement.
Step 5	Add all questions exactly as listed in this guide.
Step 6	Set all objective questions as Required. Set the final open-ended paragraph question as Required only if the supervisor wants complete feedback; otherwise it may be left not required. The questionnaire itself is still required for the project.
Step 7	Open Settings and configure: Collect email addresses Off unless supervisor requires it; Limit to 1 response Off unless using MMU accounts only; Allow response editing Off; Show progress bar On; Shuffle question order Off; Make this a quiz Off.
Step 8	Click Preview and test the form.
Step 9	Send the form link to target respondents.
Step 10	After collecting responses, export responses to Google Sheets or CSV for Chapter 3 analysis.

Section A - Respondent Background

Question	Q1. What is your age group?
Type	Multiple choice
Required	Yes
Options / Answer	Under 18; 18-25; 26-35; 36 and above

Question	Q2. What best describes you?
Type	Multiple choice
Required	Yes
Options / Answer	Software Engineering / Computer Science student; AI / Machine Learning student; Biomedical Engineering / Health-related student; Researcher; Lecturer / Supervisor; Other

Question	Q3. How familiar are you with audio processing or signal processing?
Type	Multiple choice
Required	Yes
Options / Answer	Not familiar; Basic; Intermediate; Advanced

Question	Q4. Have you worked with biomedical audio before?
Type	Multiple choice
Required	Yes
Options / Answer	Yes; No; Not sure

Question	Q5. Which type of audio-related task are you most familiar with?
Type	Multiple choice
Required	Yes
Options / Answer	Audio recording; Audio editing; Audio classification; Audio separation; Machine learning model testing; I have not worked with audio tasks before

Section B - Current Challenges in Cardiopulmonary Audio Processing

Question	Q6. How difficult do you think it is to separate heart and lung sounds from a mixed cardiopulmonary recording?
Type	Multiple choice
Required	Yes
Options / Answer	Very easy; Easy; Neutral; Difficult; Very difficult
Question	Q7. What problems do you think affect cardiopulmonary sound processing the most?
Type	Checkboxes
Required	Yes
Options / Answer	Background noise; Overlap between heart and lung sounds; Low recording quality; Difficulty identifying useful sound components; Lack of simple software tools; Lack of clear output visualization; Other
Question	Q8. Do you think noisy audio recordings can reduce the quality of machine learning-based sound processing?
Type	Linear scale or Likert multiple choice
Required	Yes
Options / Answer	Strongly disagree; Disagree; Neutral; Agree; Strongly agree
Question	Q9. Do you think a simple software prototype for heart-lung sound separation would be useful for students or researchers?
Type	Linear scale or Likert multiple choice
Required	Yes
Options / Answer	Strongly disagree; Disagree; Neutral; Agree; Strongly agree

Section C - System Feature Preferences

Question	Q10. Which feature would be most important in the proposed system?
Type	Multiple choice
Required	Yes
Options / Answer	Upload mixed cardiopulmonary audio; Preprocess noisy audio; Separate heart and lung sounds; Preview separated audio; Download separated outputs; View processing history; View evaluation results
Question	Q11. Which output would you prefer after sound separation?
Type	Multiple choice
Required	Yes
Options / Answer	Heart and lung audio files only; Audio files with waveform visualization; Audio files with spectrogram visualization; Audio files with evaluation metrics; All of the above
Question	Q12. How important is it for the system to provide a preview of the separated heart and lung sounds?
Type	Linear scale or Likert multiple choice
Required	Yes
Options / Answer	Not important; Slightly important; Moderately important; Important; Very important
Question	Q13. How important is it for the system to allow users to download separated audio outputs?
Type	Linear scale or Likert multiple choice
Required	Yes
Options / Answer	Not important; Slightly important; Moderately important; Important; Very important
Question	Q14. How important is it for the system to show basic processing information, such as file name, processing status, and completion time?
Type	Linear scale or Likert multiple choice
Required	Yes
Options / Answer	Not important; Slightly important; Moderately important; Important; Very important
Question	Q15. Should the system include a history page showing previously uploaded files and separation results?
Type	Multiple choice
Required	Yes
Options / Answer	Yes; No; Not sure

Section D - Interface and Output Preferences

Question	Q16. How would you prefer the separated results to be displayed?
Type	Multiple choice
Required	Yes
Options / Answer	Simple audio player only; Audio player with waveform; Audio player with spectrogram; Audio player with waveform, spectrogram, and metrics; Not sure
Question	Q17. Which interface style would you prefer?
Type	Multiple choice
Required	Yes
Options / Answer	Simple interface with only main features; Detailed interface with many technical options; Balanced interface with simple view and optional advanced details
Question	Q18. How important is it for the system interface to be easy for non-expert users?
Type	Linear scale or Likert multiple choice
Required	Yes
Options / Answer	Not important; Slightly important; Moderately important; Important; Very important
Question	Q19. What information should appear first after separation is completed?
Type	Multiple choice
Required	Yes
Options / Answer	Download buttons for heart and lung sounds; Audio preview players; Waveform or spectrogram visualization; Evaluation metrics; Processing status summary
Question	Q20. What type of error message would be most helpful if the uploaded file is invalid?
Type	Multiple choice
Required	Yes
Options / Answer	Short message only; Message explaining the problem; Message explaining the problem and suggested fix; Not sure

Section E - Evaluation and Improvement

Question	Q21. How important is it for the system to show evaluation metrics for the separated output?
Type	Linear scale or Likert multiple choice
Required	Yes
Options / Answer	Not important; Slightly important; Moderately important; Important; Very important

Question	Q22. Which evaluation information would be most useful?
Type	Multiple choice
Required	Yes
Options / Answer	Signal quality score; Separation accuracy score; Noise reduction information; Processing time; All of the above

Question	Q23. Would you use a prototype that separates mixed cardiopulmonary audio into heart and lung sound outputs?
Type	Multiple choice
Required	Yes
Options / Answer	Yes; No; Maybe

Question	Q24. What improvement or feature would you like to see in a cardiopulmonary sound separation system?
Type	Paragraph answer
Required	No unless supervisor requires complete written feedback
Options / Answer	Open-ended written response

After Response Collection

After real responses are collected, export the Google Form responses to Google Sheets or CSV. Use the real response data to replace the Chapter 3 placeholder figures, summarize findings, and refine the functional, non-functional, and user requirements. Do not insert percentages, charts, or findings until real response data exists.

End of guide.